

Part 3

Inferences about μ

- One of the major objectives of statistics is to make inferences about the distribution of the elements in a *population* based on information contained in a *sample*.
- Numerical summaries that characterize the population distribution are called *parameters*.
- The *population mean* μ and *population variance* σ^2 are two important parameters. Others are median, range, mode, etc.

Inferences about μ - cont.

- Methods for making inferences are basically designed to answer one of two types of questions:
 - Approximately what is the value of the parameter?
or
 - Is the value of the parameter less than (say) 6?
- Statisticians answer the first question by estimating the parameter using the sample.
- The second case might require a test of a hypothesis.

Point and Interval Estimation of μ when σ is known and n is large

- Point estimation of μ does not require that σ be known or a large n
- The point estimate of μ is the sample mean $\bar{y}$
- Point estimates by themselves do not tell how much $\bar{y}$ might differ from μ , that is, the accuracy or precision of the estimate
- A measure of accuracy is the difference between sample mean $\bar{y}$ and population mean μ is called sampling error

A Confidence Interval for μ

- An interval estimate, called a *confidence interval*, incorporates information about the amount of sampling error in $\bar{y}$
- A confidence interval for μ takes the form

$$(\bar{y} - E, \bar{y} + E),$$

for a number E

- An associated number called the *confidence coefficient* helps assess how likely it is for μ to be in the interval.

A Confidence Interval for μ - cont.

- To derive the specific form of the confidence interval for μ , for the case when σ is known and n is large, the CLT result must be used.
- By the CLT, $Z = (\bar{Y} - \mu)/\sigma_{\bar{y}}$ is approximately $\sim N(0, 1)$.
- Let Z have a $N(0, 1)$ distribution (exactly).
- Let $z_{\alpha/2}$ denote the $1 - \alpha/2$ quantile of the standard normal distribution, for a given number α , $0 < \alpha < 1$.
- Then the following probability statement is true:

$$P\left(-z_{\alpha/2} \leq \frac{(\bar{Y} - \mu)}{\sigma_{\bar{y}}} \leq z_{\alpha/2}\right) = 1 - \alpha$$

A Confidence Interval for μ - cont.

- Manipulating the inequalities, without changing values, we have

$$P(\sigma_{\bar{y}} z_{\alpha/2} \geq \mu - \bar{Y} \geq -\sigma_{\bar{y}} z_{\alpha/2}) = 1 - \alpha$$

$$P(\bar{Y} - \sigma_{\bar{y}} z_{\alpha/2} \leq \mu \leq \bar{Y} + \sigma_{\bar{y}} z_{\alpha/2}) = 1 - \alpha$$

- If, for example, $\alpha = 0.05$ then

$$P(\bar{Y} - \frac{\sigma}{\sqrt{n}} z_{.025} \leq \mu \leq \bar{Y} + \frac{\sigma}{\sqrt{n}} z_{.025}) = 0.95$$

and, since $z_{0.025} = 1.96$,

$$P(\bar{Y} - 1.96 \frac{\sigma}{\sqrt{n}} \leq \mu \leq \bar{Y} + 1.96 \frac{\sigma}{\sqrt{n}}) = 0.95$$

A Confidence Interval for μ - cont.

- Using this statement, a $(1-\alpha)100\%$ confidence interval for μ , with confidence coefficient $1 - \alpha$ can be calculated using a random sample of data $y_1, y_2, \dots, y_n$
- It is usually written in the form

$$\left(\bar{y} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, \bar{y} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2} \right).$$

- **Example:** Suppose $n = 36$, $\sigma = 12$, $\bar{y} = 24.8$
And for $\alpha = 0.05$, $z_{\alpha/2} \equiv z_{0.025} = 1.96$

A Confidence Interval for μ - cont.

- Thus a 95% C.I. for μ is

$$\left(24.8 - \frac{12}{\sqrt{36}} \times 1.96, 24.8 + \frac{12}{\sqrt{36}} \times 1.96 \right)$$

- The interval for μ is: (20.88, 28.72) with confidence coefficient 0.95.
- We might say that we are 95% *confident* that the population mean μ is between 20.88 and 28.72.
- But what do we actually mean when we say that we are 95% *confident*?

Interpretation of a Confidence Interval

- Before the sample is drawn, the probability is $(1 - \alpha)$ that the random interval $(\bar{Y} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, \bar{Y} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2})$ will contain μ (because $\bar{Y}$ is a random variable).
- However, once the sample is drawn, and $\bar{y}$ is calculated, the interval ceases to be random. It is a numerical interval $(\bar{y} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, \bar{y} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2})$, calculated specifically for the drawn sample; thus we cannot associate a probability with it.
- We may say that the process which led us to this interval will, on the average, produce an interval containing μ , $100(1 - \alpha)\%$ of the time.

Interpretation of a Confidence Interval - cont.

- *It is not correct to say that the interval $(\bar{y} - \frac{\sigma}{\sqrt{n}} z_{\alpha/2}, \bar{y} + \frac{\sigma}{\sqrt{n}} z_{\alpha/2})$ contains μ with a specified probability.*
- If the process of sampling is repeated many times and $100(1-\alpha)\%$ confidence intervals calculated for each sample, then we are confident that $100(1-\alpha)\%$ of those intervals will contain μ .

Notes:

- As the confidence coefficient increases the confidence interval becomes wider. A wider interval estimates μ less precisely. Thus a 99% confidence interval is less accurate than a 95% confidence interval but one has more confidence that μ is in the first interval.
- Also note that increasing the sample size n results in a narrower interval (more accurate estimate) for the same confidence coefficient.

Example:

The caffeine content (in mg) was examined for a random sample of 50 cups of black coffee dispensed by a new machine. The mean and standard deviation were 110 mg and 7.1 mg, respectively. Use these data to construct a 98% confidence interval for μ , the mean caffeine content for cups dispensed by the machine.

As the sample size is large we can use the CLT and also approximate σ , the sample standard deviation of the population by $s = 7.1$

Example - cont.

$$\left[\bar{y} - z_{0.01} \frac{\sigma}{\sqrt{n}}, \bar{y} + z_{0.01} \frac{\sigma}{\sqrt{n}} \right]$$
$$\left[110 - (2.33) \left(\frac{7.1}{\sqrt{50}} \right), 110 + (2.33) \left(\frac{7.1}{\sqrt{50}} \right) \right]$$
$$(107.66, 112.34)$$

We are 98% confident that the mean caffeine content for cups dispensed by the machine is between 107.66 and 112.34 mg.

Choosing the Sample Size

- The width of a confidence interval is $(2 \times \sigma \times z_{\alpha/2})/\sqrt{n}$.
- The width of the confidence interval can be made smaller by changing to a larger α or increasing sample size n .
- Let us consider selection of n to achieve a desired width $2 \cdot E$ for a fixed α
- We want E to at least equal to $\sigma \times z_{\alpha/2}/\sqrt{n}$
- Thus we need to select a sample so that

$$n \geq \frac{(z_{\alpha/2})^2 \sigma^2}{E^2}$$

Choosing the Sample Size - cont.

- This will result in an interval no wider than $(\bar{y} - E, \bar{y} + E)$.

Example:

Given $\sigma = 12$, $\alpha = 0.05$, $z_{0.025} = 1.96$

What sample size n will give an interval no wider than 5.6?

We set $E = 2.8$ so

$$n \geq \frac{(1.96)^2 \cdot 144}{(2.8)^2} = 70.56$$

Thus the experimenter must choose a sample size $n = 71$ at least.

Choosing the Sample Size - cont.

Would the sample size $n = 71$ ensure the width to be ≤ 5.6 if the population has a larger variance? Would $n = 71$ be enough?

The answer is **NO** since the formula

$$\frac{(z_{\alpha/2})^2 \sigma^2}{E^2}$$

involves the population variance σ^2 .

Precision in estimation depends on both α and σ^2 . If the variance of population elements is very small, i.e., the elements are tightly clustered about the population mean μ , then only a small sample is needed for the estimate $\bar{y}$ to be very near μ .

Statistical Tests for μ

- Estimation (either point or interval estimation) was used to help answer a question like “Approximately what is the value of μ ?”
- The other kind of question mentioned earlier is “Is it likely that μ is less than (or greater than) the value μ_0 (a pre-determined value)?”.
- A Test of Hypothesis is used to answer this kind of question. As in estimation, the sample mean $\bar{y}$ of a random sample of n elements from the population is used to answer this question.

Statistical Tests for μ - cont.

- Every test of hypothesis features
 - a Null Hypothesis H_0 which describes a characteristic of the population (it is believed to be) as it currently exists,
 - a Research Hypothesis (or Alternative Hypothesis) H_a , which is a proposal about this characteristic, by the person(s) conducting the statistical study.
- The idea is that the null hypothesis is presumed to hold unless there is overwhelming evidence in the data in support of the research hypothesis.

Example:

- To determine whether the mean yield per acre (in bushels), μ , of a variety of soybeans increased the current year over the last two years when μ is believed to be 520 bushels per acre, the following might be tested.

$$H_0 : \mu \leq 520 \quad \text{vs.} \quad H_a : \mu > 520$$

- The fact that μ may equal a specific value is always included in the null hypothesis. The decision to state whether the data supports the research hypothesis or not is based on a quantity computed from the data called the test statistic.